Architecture Specification v1.0

Parameterized INT8/FP16 Systolic Tensor Core with Online Softmax Unit

Target: Arty A7-100T (XC7A100T) — FCCM 2027 Reconfigurable Computing Challenge

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Project Goal

Design and implement a fully parameterized, synthesizable systolic tensor core on the Xilinx XC7A100T FPGA that accelerates INT8 matrix multiplication with FP32 accumulation and a streaming online softmax unit, targeting the attention score computation $QK^\top$ in transformer self-attention. The design produces FP16 outputs, with FP32 used exclusively for intermediate computation. The summer 2026 deliverable is a verified, timing-closed N=8 prototype. The April 2027 submission extends to N=16 via tiling. All RTL is parameterized with `N` as a `generate` parameter.

System Architecture

The top-level design is a four-stage feed-forward pipeline. BRAM ping-pong buffers supply Q and K tiles to a skew network, which feeds the $N \times N$ systolic array. Each PE output passes through a dedicated INT32→FP32 conversion unit, and the resulting FP32 tile streams into the online softmax unit.

The systolic array computes $C = QK^\top$ tile by tile, which is the core operation of transformer attention as introduced in [1]. This design follows recent work on fusing attention computation into systolic arrays [2], enabling elimination of external vector processing units.

Dataflow: Output-Stationary

The systolic array uses **output-stationary** dataflow. Weight-stationary is not suitable for $QK^\top$ because K is a runtime activation, not a persistent weight tensor. Keeping partial sums local avoids repeated FP accumulation reordering effects, ensuring only a single FP32 rounding event per output activation.

Tile Size: 8

The RTL is parameterized with `N`. The baseline uses **N=8** (64 PEs, 64 DSP48E1 slices). Larger matrices are handled via tiling.

Tiling preserves compute density but increases memory traffic. Correctness requires that all tiles contributing to a given output row are processed sequentially, enabling row-wise softmax state preservation.

Precision and Scaling

The compute pipeline is:

$$\text{INT8} \times \text{INT8} \rightarrow \text{INT32} \rightarrow \text{FP32} \rightarrow \text{FP16}$$

The INT32 accumulator resides in DSP48E1 [3]. After accumulation, values are converted to FP32 using a LUT-based unit.

Attention scaling factor: The Transformer introduces a normalization factor of $1/\sqrt{d_k}$ in dot-product attention [1]. This factor is fused into the INT32→FP32 conversion stage to avoid additional multiply latency and to reduce rounding events.

Final output format: FP16 (stored after softmax normalization).

Rounding mode: Round-to-Nearest Ties-to-Even (RNTE), IEEE 754 default [4].

BRAM and Bandwidth

Bandwidth is the primary constraint. The array consumes N values per cycle.

Ping-pong buffering overlaps:

- DMA loading of next tile
- systolic computation of current tile

This ensures no structural memory hazards. BRAM constraints are derived from device limits [5].

Latency Model

For an $N \times N$ systolic array, output timing follows the wavefront propagation model [6]:

$$t_{i,j} = i + j + N \tag{1}$$

Worst-case latency:

$$T_{\text{array}} = 3N - 2 \tag{2}$$

DSP pipeline latency adds 4[3]:

$$T_{\text{tile}} = (3N - 2) + 4 \tag{3}$$

For N=8: 26 cycles.

Streaming Online Softmax

Softmax uses the numerically stable online formulation [7]:

$$\text{softmax}(x_i) = \frac{e^{x_i - m}}{\sum_j e^{x_j - m}}, \quad m = \max(x) \tag{4}$$

Key property: Row-wise softmax is stateful and must preserve:

- running maximum m
- running normalization sum ℓ

This introduces a loop-carried dependency with II = 1.

exp() Implementation (AHBU)

The exponential function uses AHBU [8].

Input domain: $[-8, 0]$ (enforced by softmax stabilization shift)

Pipeline:

$$\text{FP32} \rightarrow \text{Fixed}\langle 1, 16, 12 \rangle \rightarrow \text{AHBU} \rightarrow \text{FP32}$$

AHBU provides improved area-latency efficiency compared to CORDIC/LUT-based approaches [8].

Numerical Considerations

The system combines:

- INT8 quantization
- FP32 accumulation
- approximate exponential evaluation

Error sources include quantization noise, rounding, and approximation error. Stability under combined error sources must be validated empirically.

Clock Target

Target clock: **200 MHz**

Critical paths:

- skew network routing
- INT32→FP32 conversion
- FP32→fixed conversion
- AHBU exp()

Final frequency determined post place-and-route.

Open Questions (May 7 Meeting)

- FP32 ↔ fixed-point interface for AHBU
- Config 1 vs Config 2 accuracy tradeoff
- AHBU latency on Artix-7 after synthesis
- combined quantization + approximation error in softmax
- cross-tile row-wise softmax correctness

This document defines the architectural baseline prior to RTL implementation.

References

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[8] A. Khataei, G. Singh, and K. Bazargan, "Approximate hybrid binary-unary computing with applications in BERT language model and image processing," pp. 1–11, 2023.